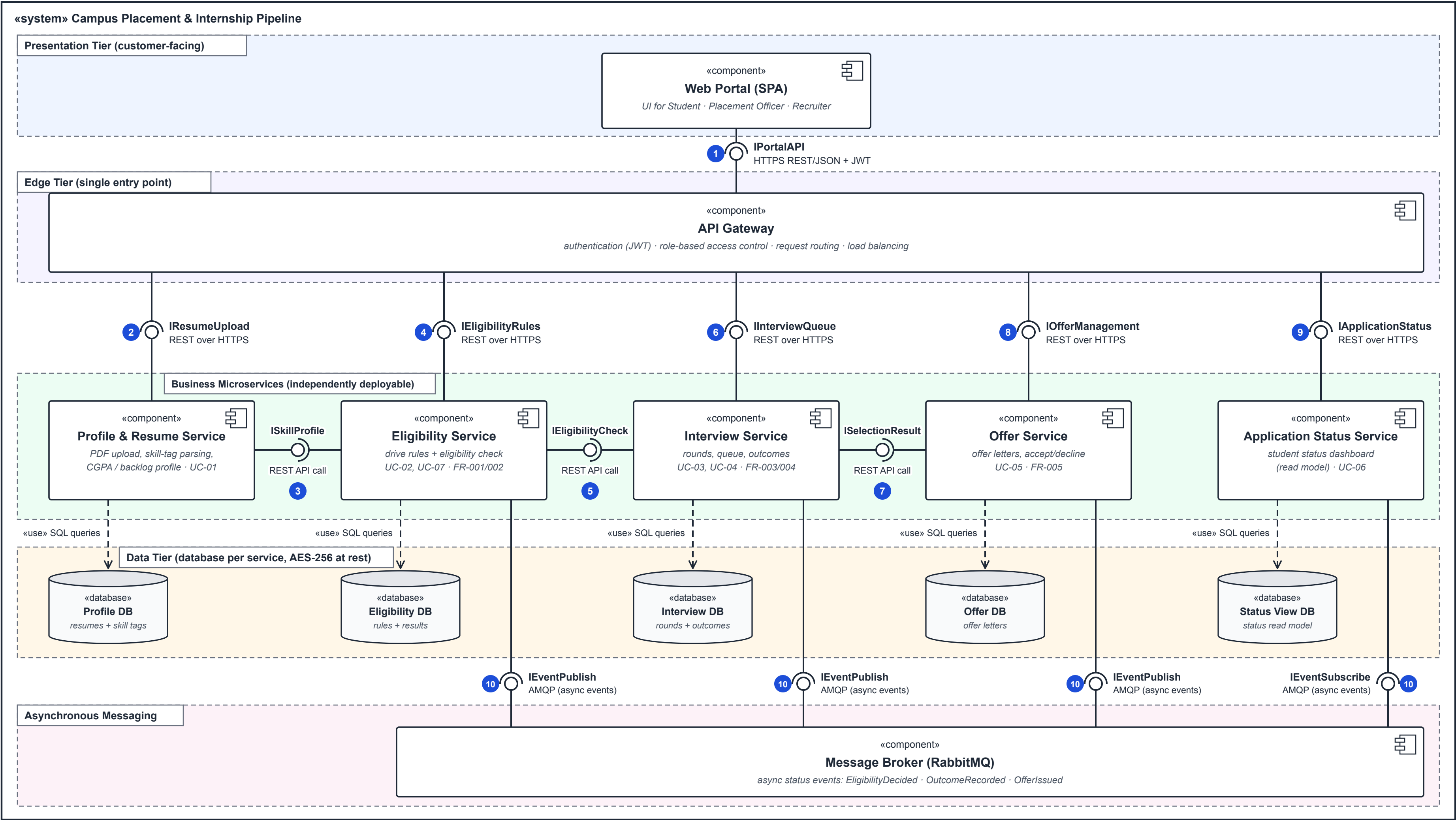


Campus Placement & Internship Pipeline – UML Component Diagram

Lab 3: Component Modelling & Architectural Pattern Selection | Selected architecture: Microservices Architecture

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Notation

-  «component»: replaceable black-box module
-  Provided interface (ball)
-  Required interface (socket)
-  Assembly connector – internal call (solid line, provided ↔ required)
-  «use» usage dependency (dashed arrow)
-  «database»: data store owned by exactly one service
-  Tier (package grouping of components)
-  Interface number (see table)
-  «system» boundary

Interfaces & data flow (provided by → required by)

#	Interface	Provider → Consumer	Technology	Operations / data exchanged
1	IPortalAPI	API Gateway → Web Portal	HTTPS REST/JSON + JWT	login(), every user request from the browser
2	IResumeUpload	Profile & Resume Service → API Gateway	REST over HTTPS	uploadResume(pdf), getProfile(studentId)
3	ISkillProfile	Profile & Resume Service → Eligibility Service	REST API call	getSkillTags(studentId), getCgpaAndBacklogs(studentId)
4	IEligibilityRules	Eligibility Service → API Gateway	REST over HTTPS	configureRules(driveId, rules), runCheck(studentId, driveId)
5	IEligibilityCheck	Eligibility Service → Interview Service	REST API call	isEligible(studentId, driveId): result + failed rules
6	IInterviewQueue	Interview Service → API Gateway	REST over HTTPS	createRound(), reorderQueue(), recordOutcome()
7	ISelectionResult	Interview Service → Offer Service	REST API call	isSelected(studentId, driveId): boolean
8	IOfferManagement	Offer Service → API Gateway	REST over HTTPS	createOffer(), uploadOfferLetter(), respondToOffer()
9	IApplicationStatus	Application Status Service → API Gateway	REST over HTTPS	getApplications(studentId), getOfferStatus(studentId)
10	IEventPublish / Subscribe	Message Broker → Eligibility, Interview, Offer / Status	AMQP (async)	EligibilityDecided, OutcomeRecorded, OfferIssued events

Data flow: a student uploads a PDF resume (1 → 2) and the Profile & Resume Service extracts skill tags. The Placement Officer configures drive rules and runs screening (4), which reads the student's skills, CGPA and backlogs (3). Only eligible students can be queued for interview rounds (6, checked via 5). Recruiter outcomes are recorded (6); an offer can be created only for a Selected applicant (8, checked via 7). Each status change is published as an event (10) into the Application Status Service's read model, which students view (9).